

Lab 9

Treble and Bass Assembly

Shaymon Madrid

Samuel Bourgeois

7/10/2025

ECEN 299

Objectives

Objectives of this session are to assemble and analyze to Treble and Bass to ensure that it conforms to given requirements and that it functions properly.

Procedure

Equipment and supplies

- Function Generator
- Oscilloscope
- Jumper wires
- PCB board and Components
- Project motherboard
- Microscope
- Soldering Iron and other needed soldering items

Procedure

For this lab, we finished assembly of the final board. We soldered our components to the Treble and bass board. Most of this report will relate more to the end processes of troubleshooting and evaluating the functionality of the board. To begin, we first analyzed the Bass channels for bot a=0 and a=1 positions for the potentiometers. We then analyzed the Treble channels for the same potentiometer positions. We only need one treble screenshot for low frequencies since we have the low frequencies being handled by the Bass channels; the same reasoning goes for the Bass at high frequencies. These both are made to affect each respectively. Table one below has our recorded and calculated values for the Adb. The screenshots are also organized by Treble and Bass below for each required image of the oscilloscope measurements.

Table 1 – Frequency Response Measurements

		Treble			Bass			
Channel	f(Hz)	Vin	Vout	Adb	Vin	Vout	Adb	
a=0 (knob left)	Left	20Hz	4.00V	4.08V	.17	4.16V	1.44V	-9.21
		200Hz	4.00V	4.00V	0	4.00V	1.44V	-8.87
		2kHz	4.00V	2.08V	-5.68	4.00V	3.52V	-1.11
		20kHz	4.08V	1.04V	-	4.08V	4.00V	-0.17
	Right	20Hz	4.00V	4.08V	0.17	4.08V	1.28V	-
		200Hz	4.00V	4.00V	0	4.08V	1.44V	10.07
		2kHz	4.00V	2.00V	-6.02	4.08V	1.44V	-9.05
		20kHz	4.00V	2.00V	-	4.08V	3.60V	-1.09
		20kHz	4.16V	1.20V	10.80	4.08V	3.92V	-0.35
		20kHz	4.16V	1.20V	10.80	4.08V	3.92V	-0.35
a=1 (knob right)	Left	20Hz	4.00V	4.00V	0	4.08V	13.3V	10.26
		200Hz	4.00V	4.08V	0.17	4.08V	11.4V	8.92
		2kHz	3.92V	7.76V	5.93	4.08V	4.56V	0.97

Right	20kHz	4.00V	15.6V	11.82	4.08V	4.00V	-0.17
	20Hz	4.00V	4.08V	0.17	4.08V	13.1V	10.13
	200Hz	4.00V	4.16V	0.34	4.08V	11.4V	8.92
	2kHz	4.00V	7.84V	5.85	4.08V	4.56V	0.97
	20kHz	4.00V	15.4V	11.71	4.08V	4.00V	-0.17

Treble Figures:

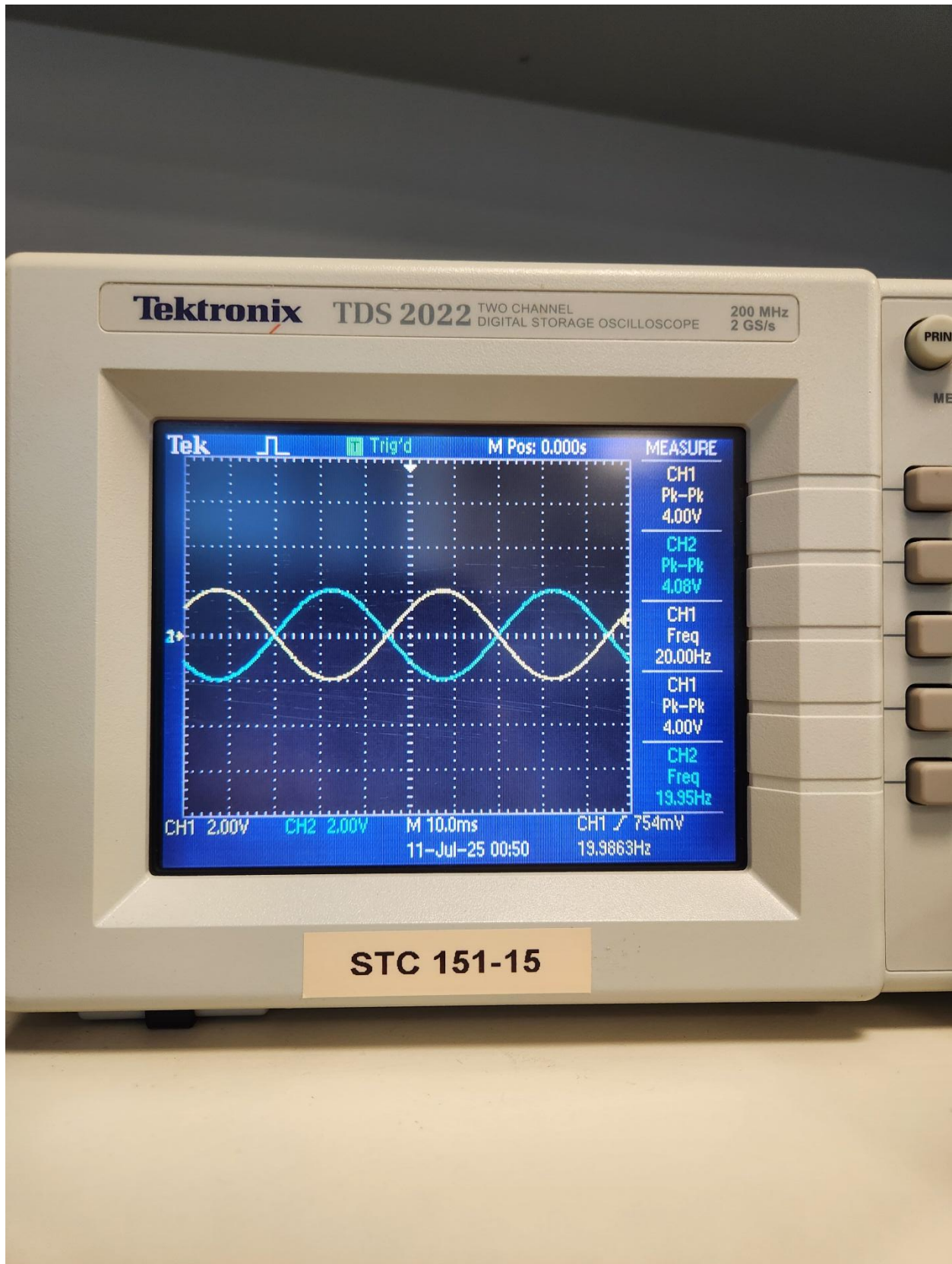


Figure 1: Treble Left (a=0, 20Hz)

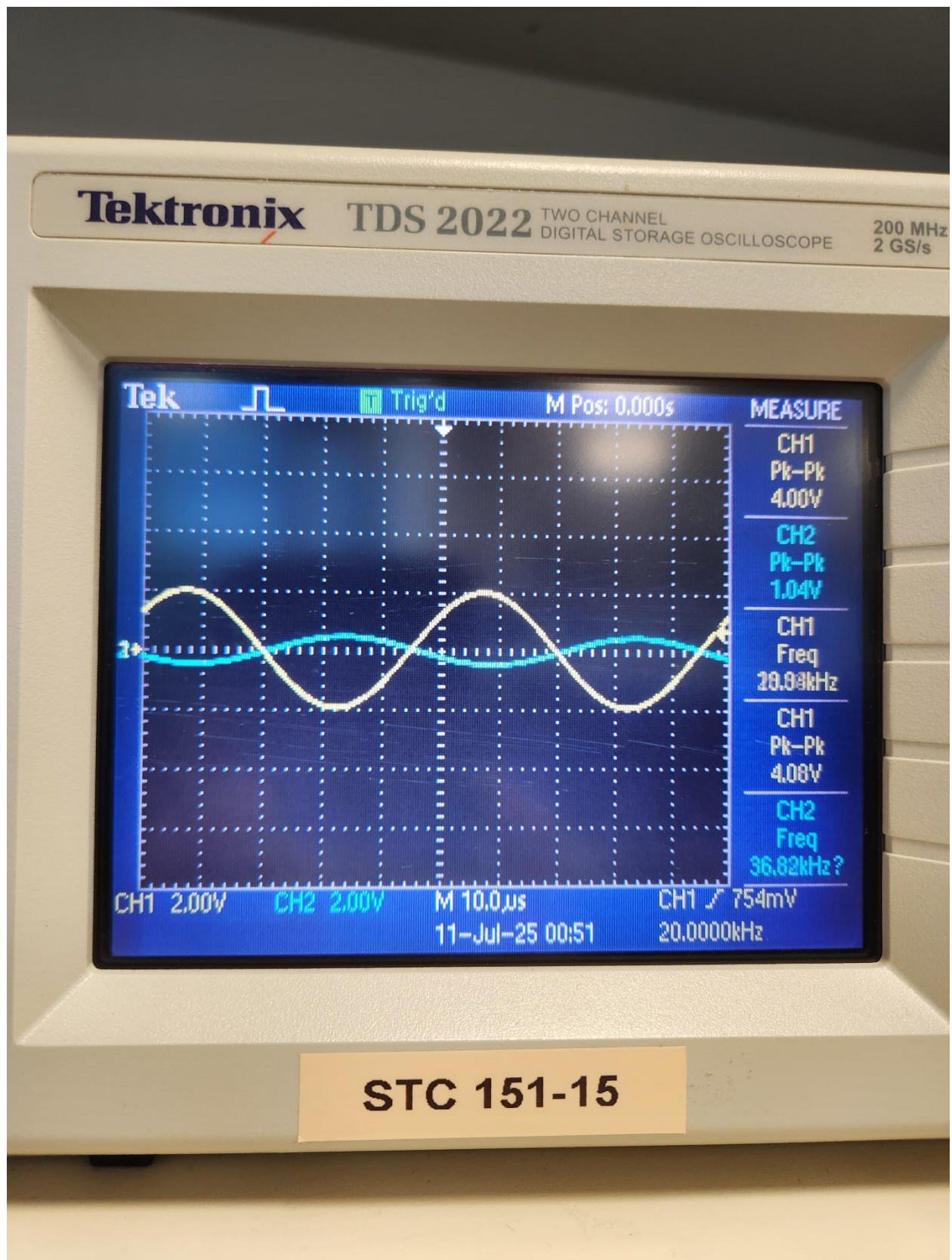


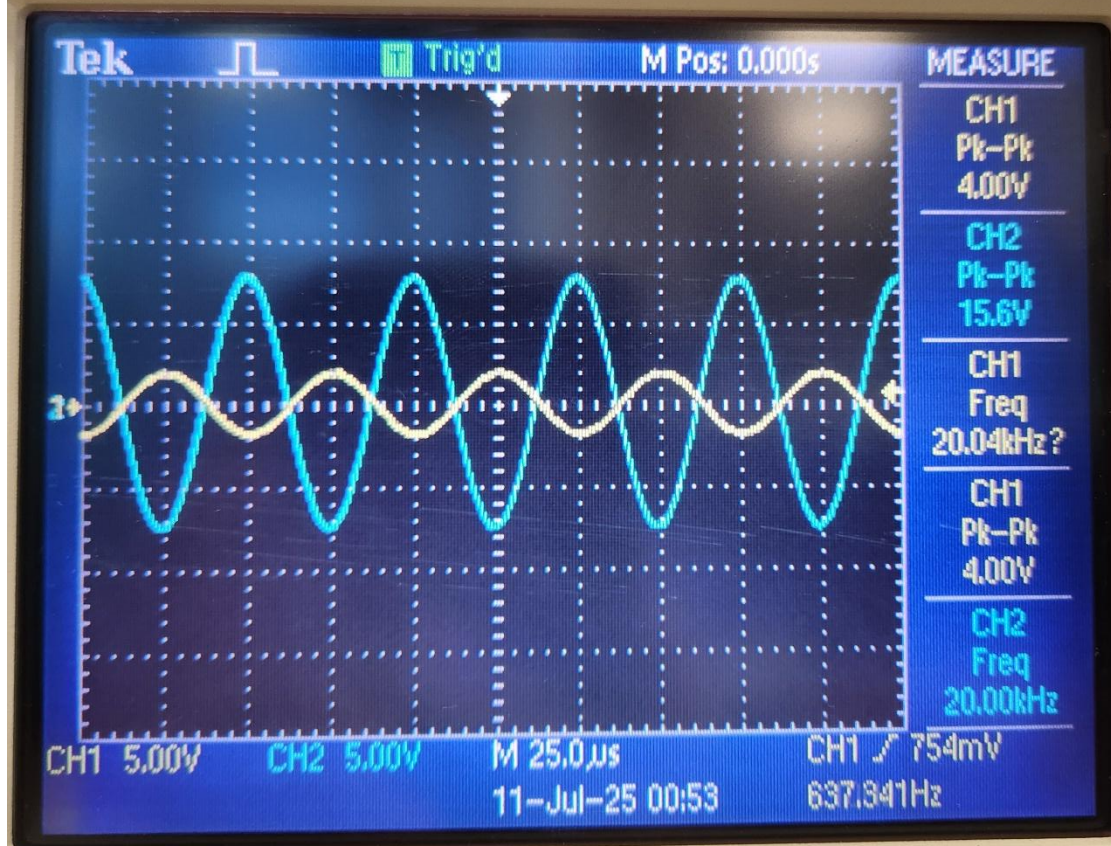
Figure 2: Treble Left ($a=0$, 20kHz)

tektronix

TDS 2022

TWO CHANNEL
DIGITAL STORAGE OSCILLOSCOPE

200 MHz
2 GS/s



STC 151-15

Figure 3: Treble Left ($a=1$, 20kHz)

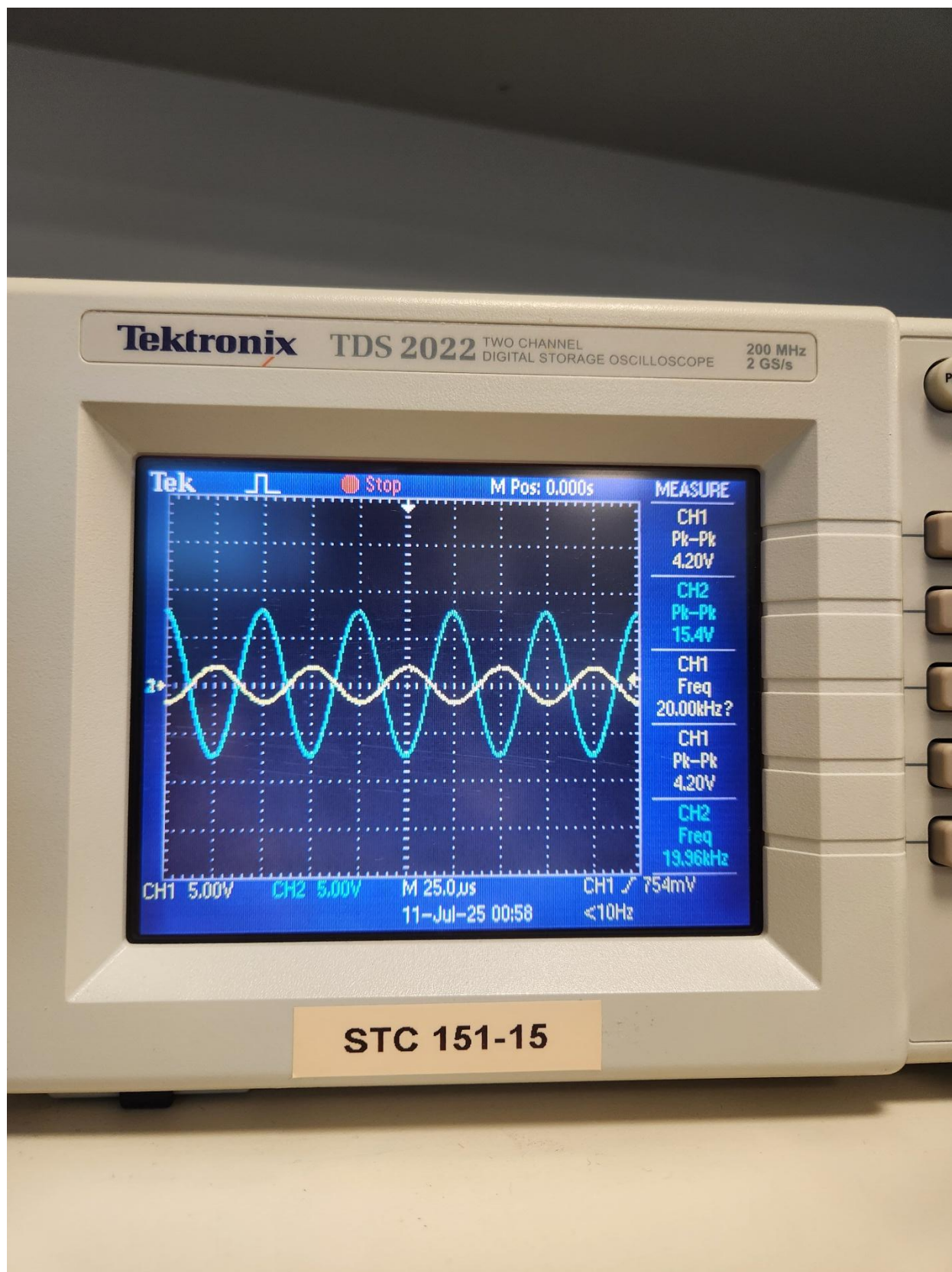


Figure 4: Treble Right ($a=1$, 20kHz)

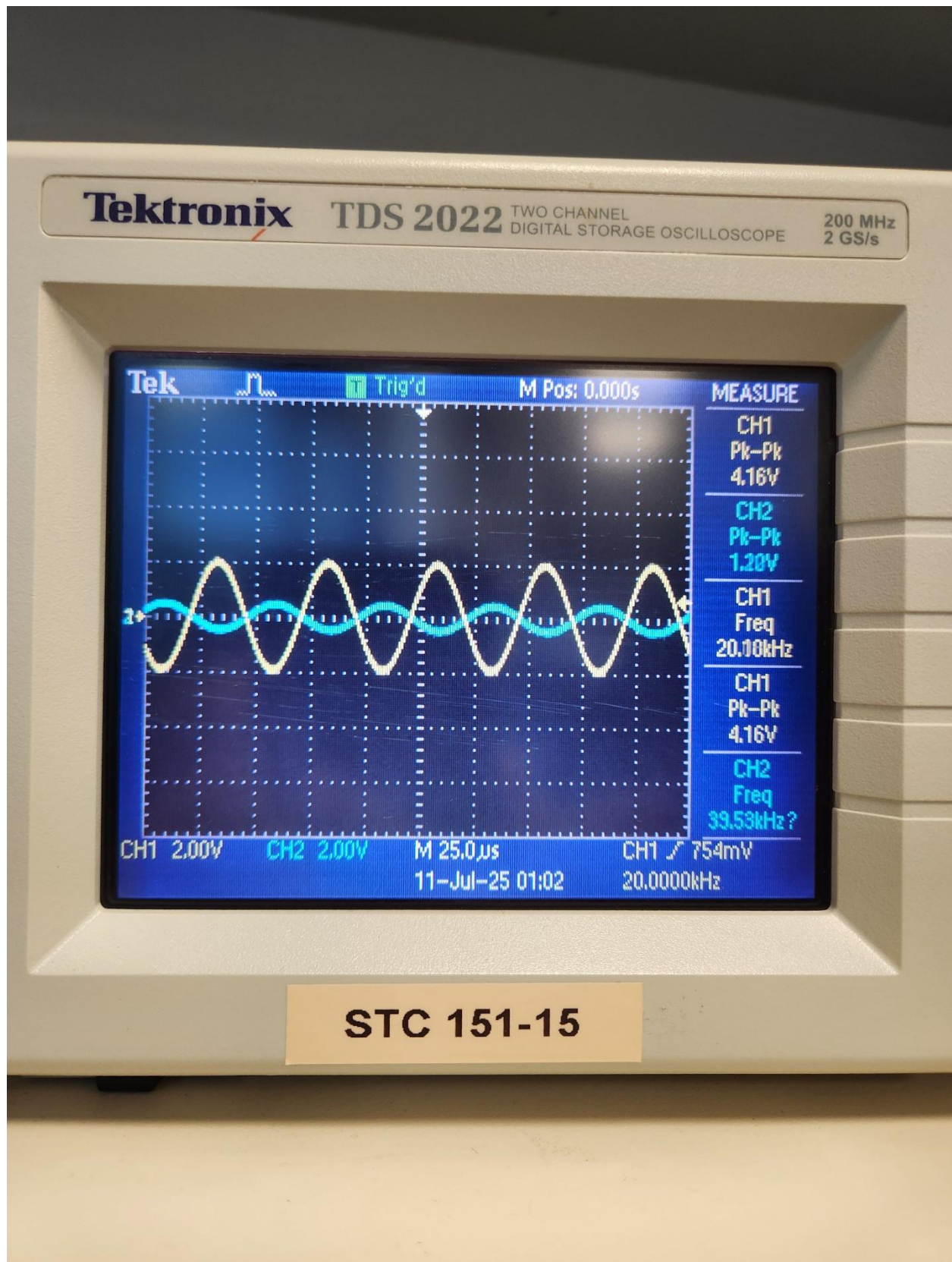


Figure 5: Treble Right ($a=0$, 20kHz)

Bass Figures:

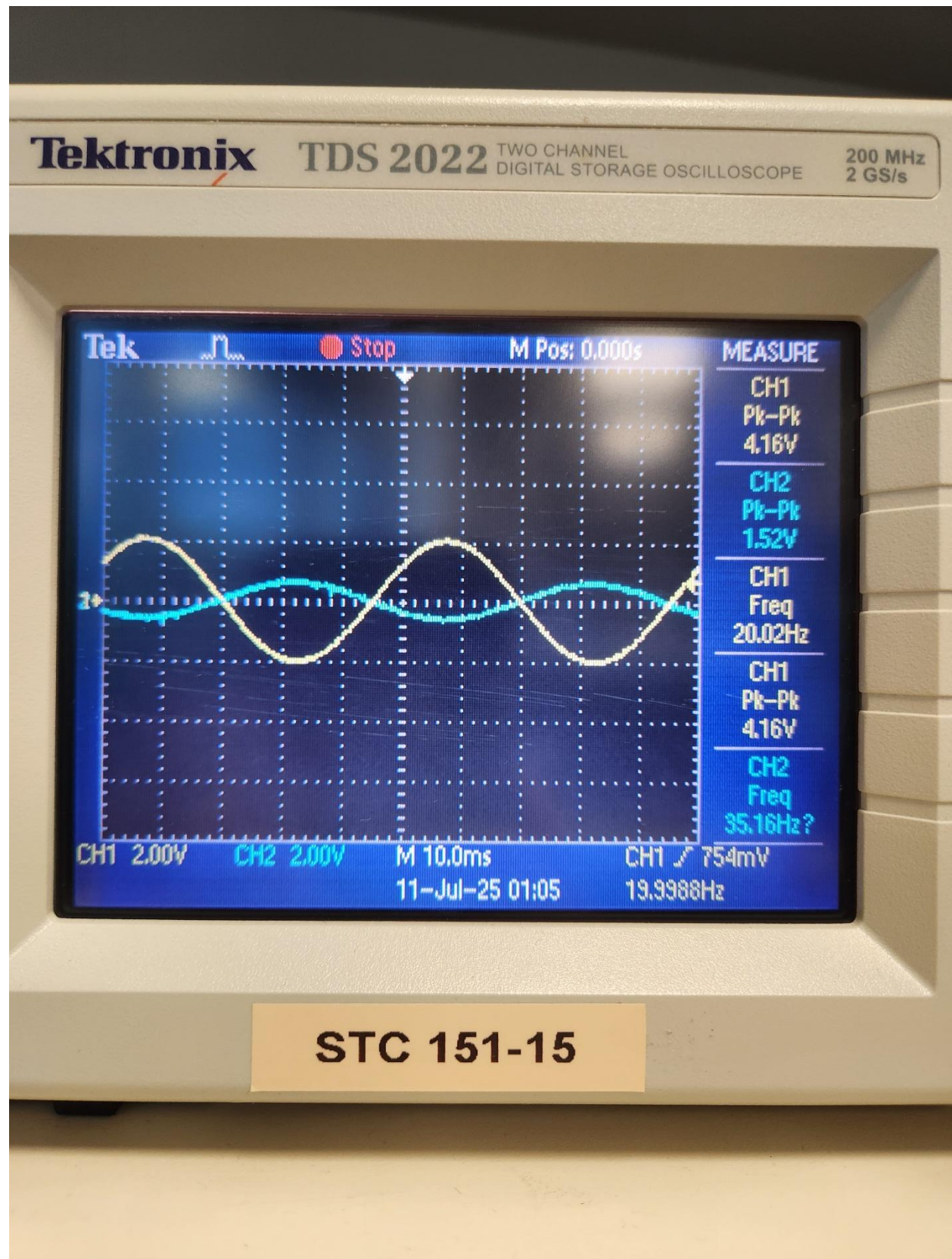


Figure 6: Bass Left ($a=0$, 20Hz)

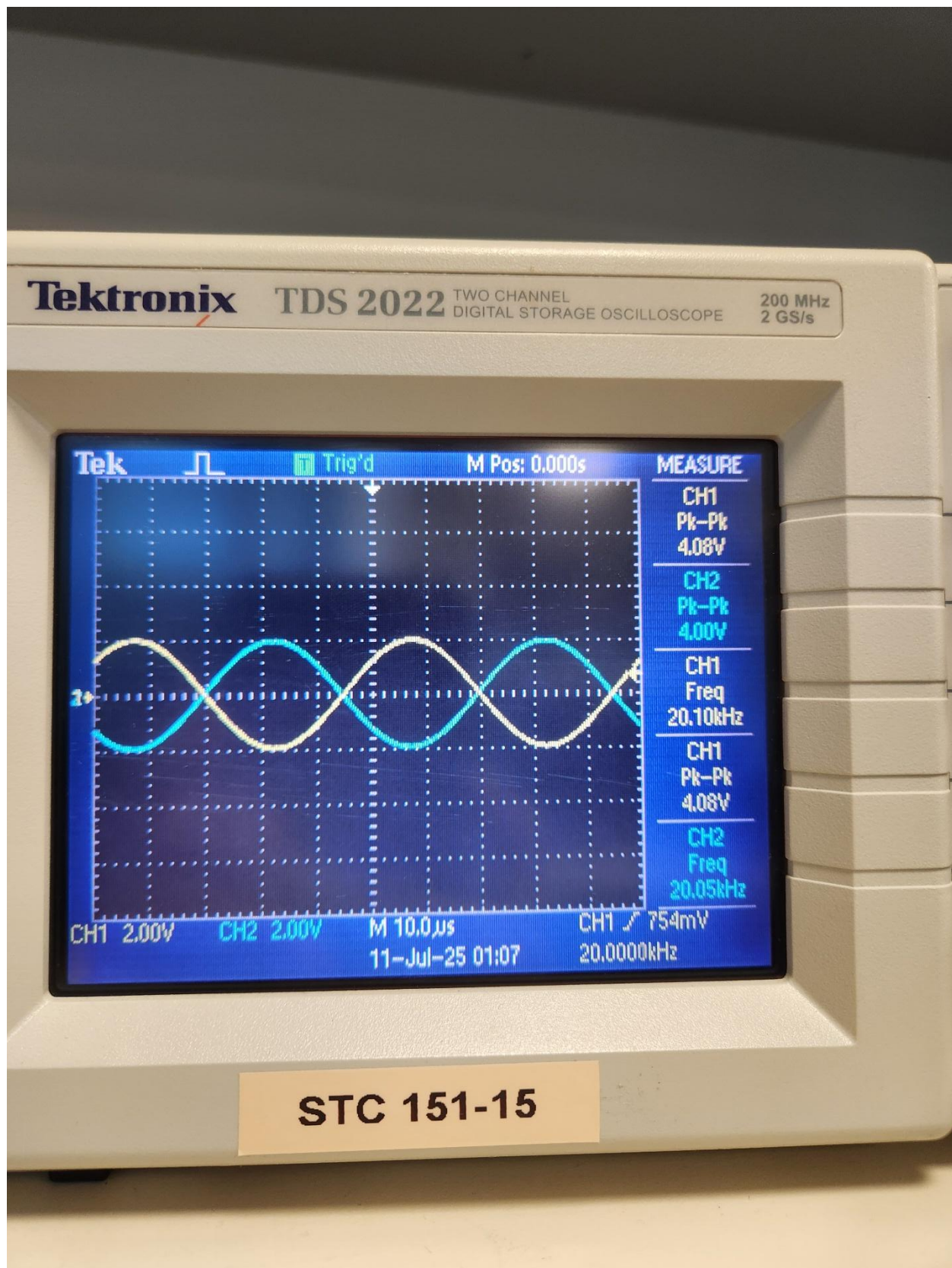
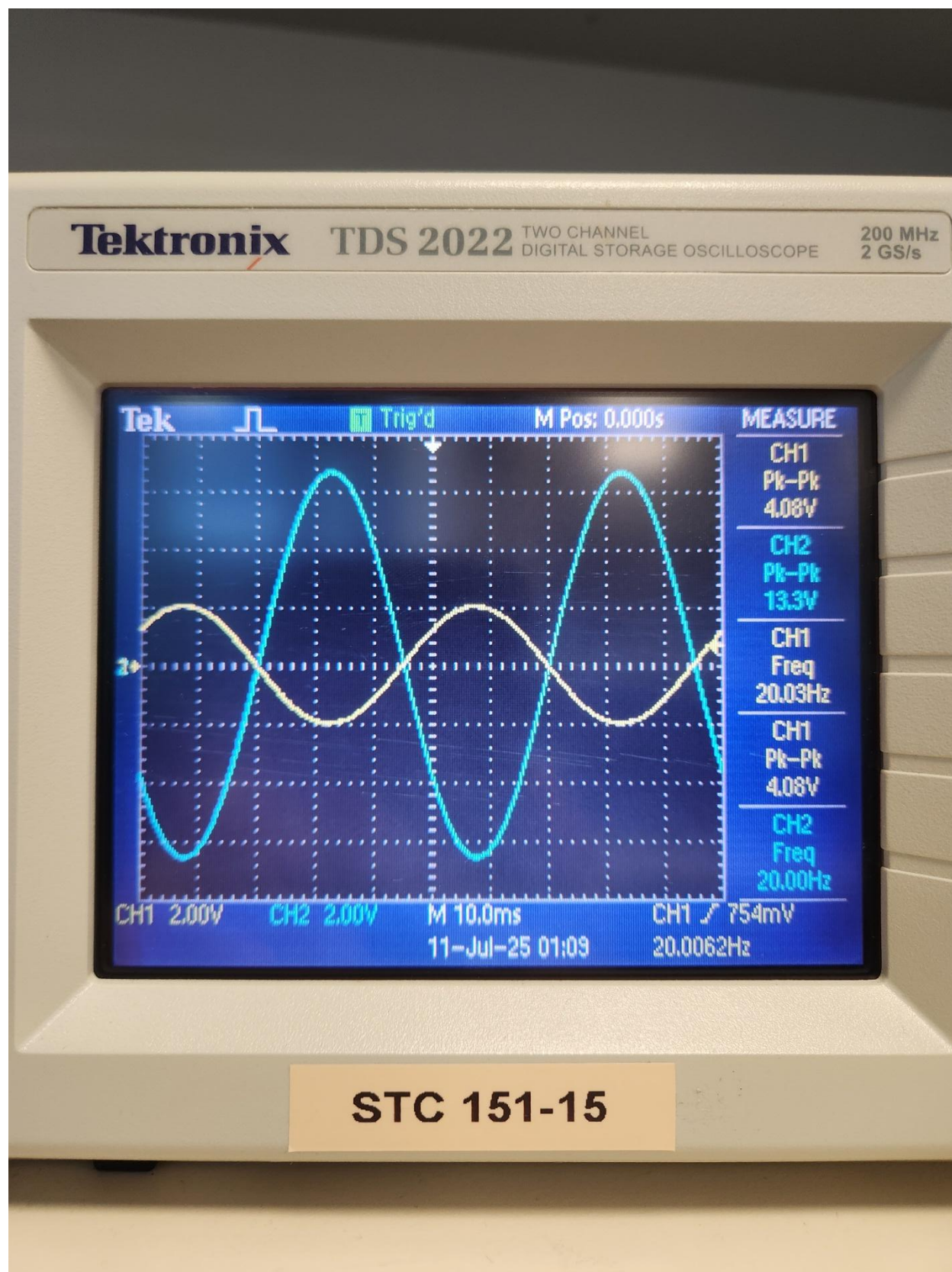


Figure 7: Bass Left ($a=0$, 20kHz)



STC 151-15

Figure 8: Bass Left ($a=1$, 20Hz)

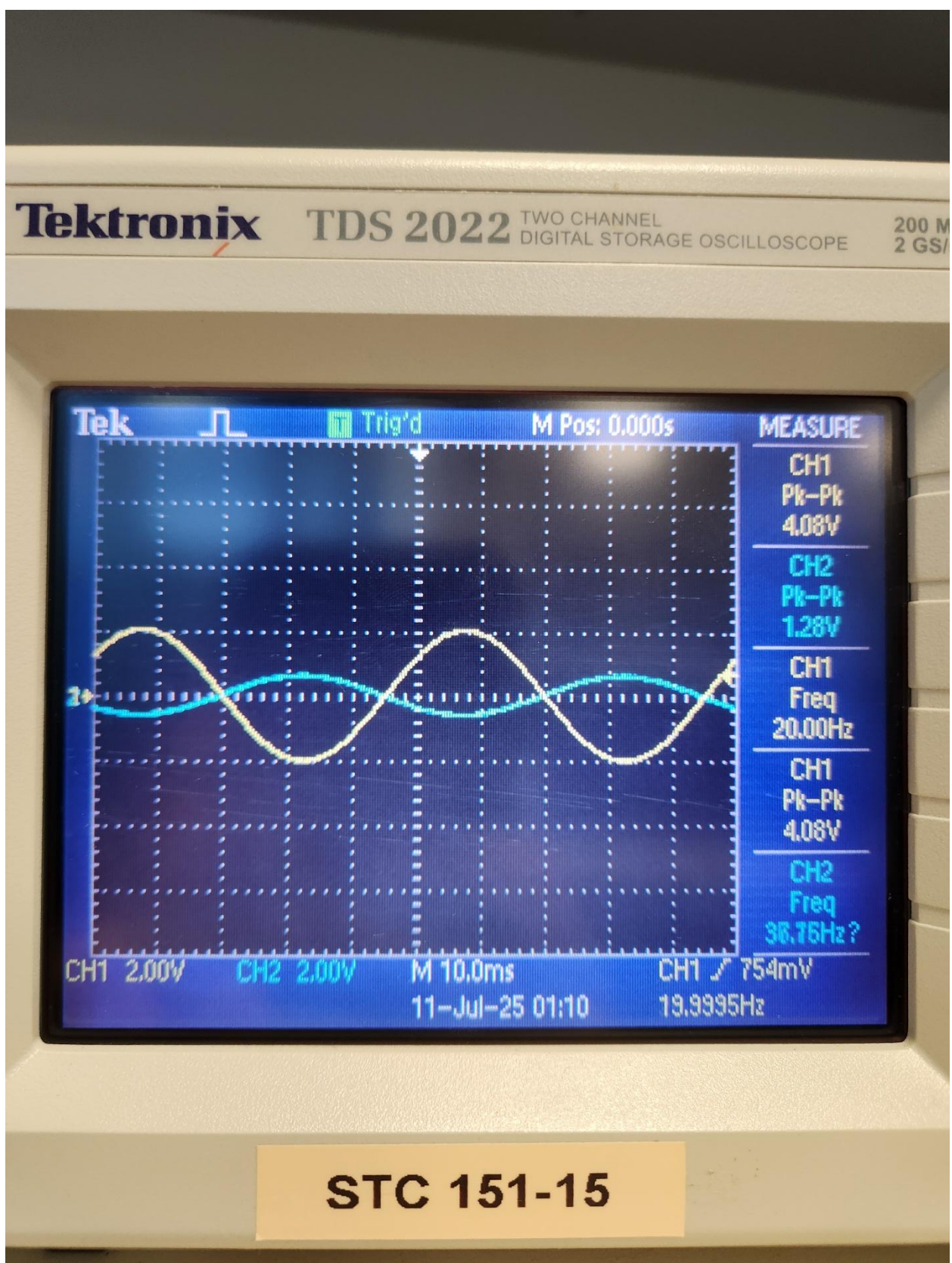


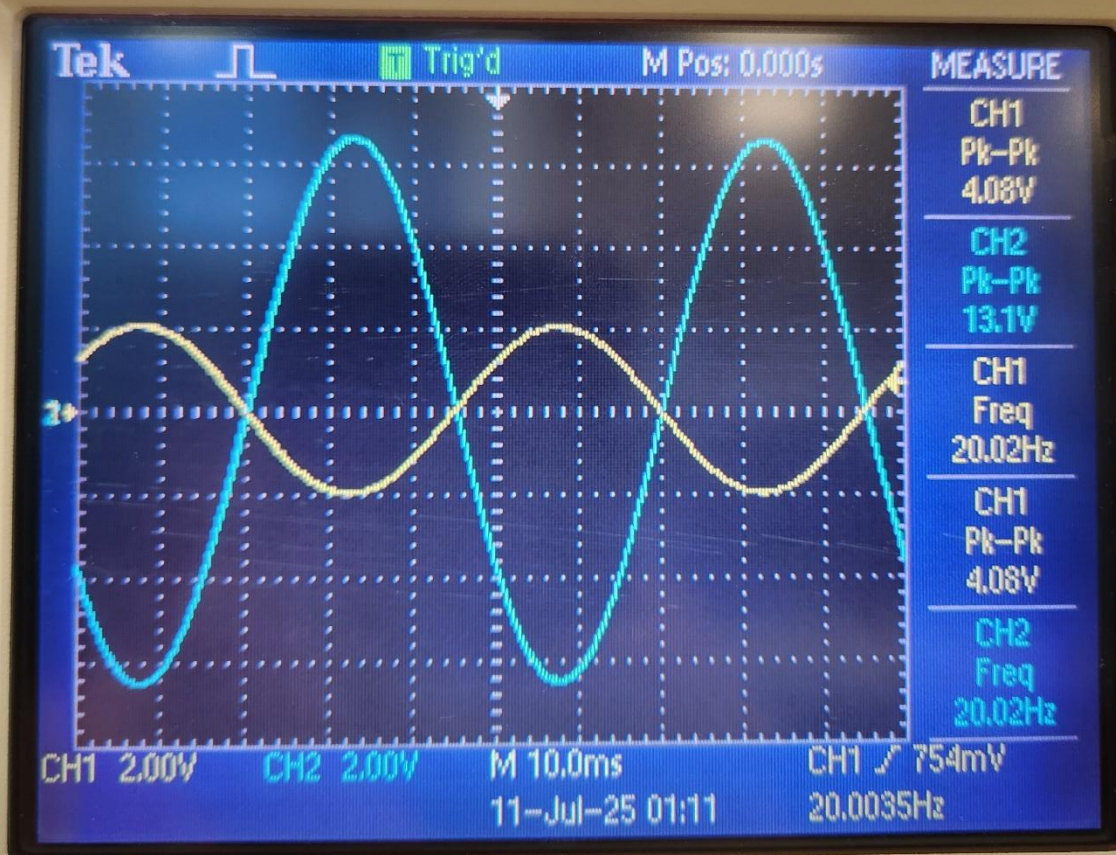
Figure 9: Bass Right ($a=0$, 20Hz)

Tektronix

TDS 2022

TWO CHANNEL
DIGITAL STORAGE OSCILLOSCOPE

200 MHz
2 GS/s



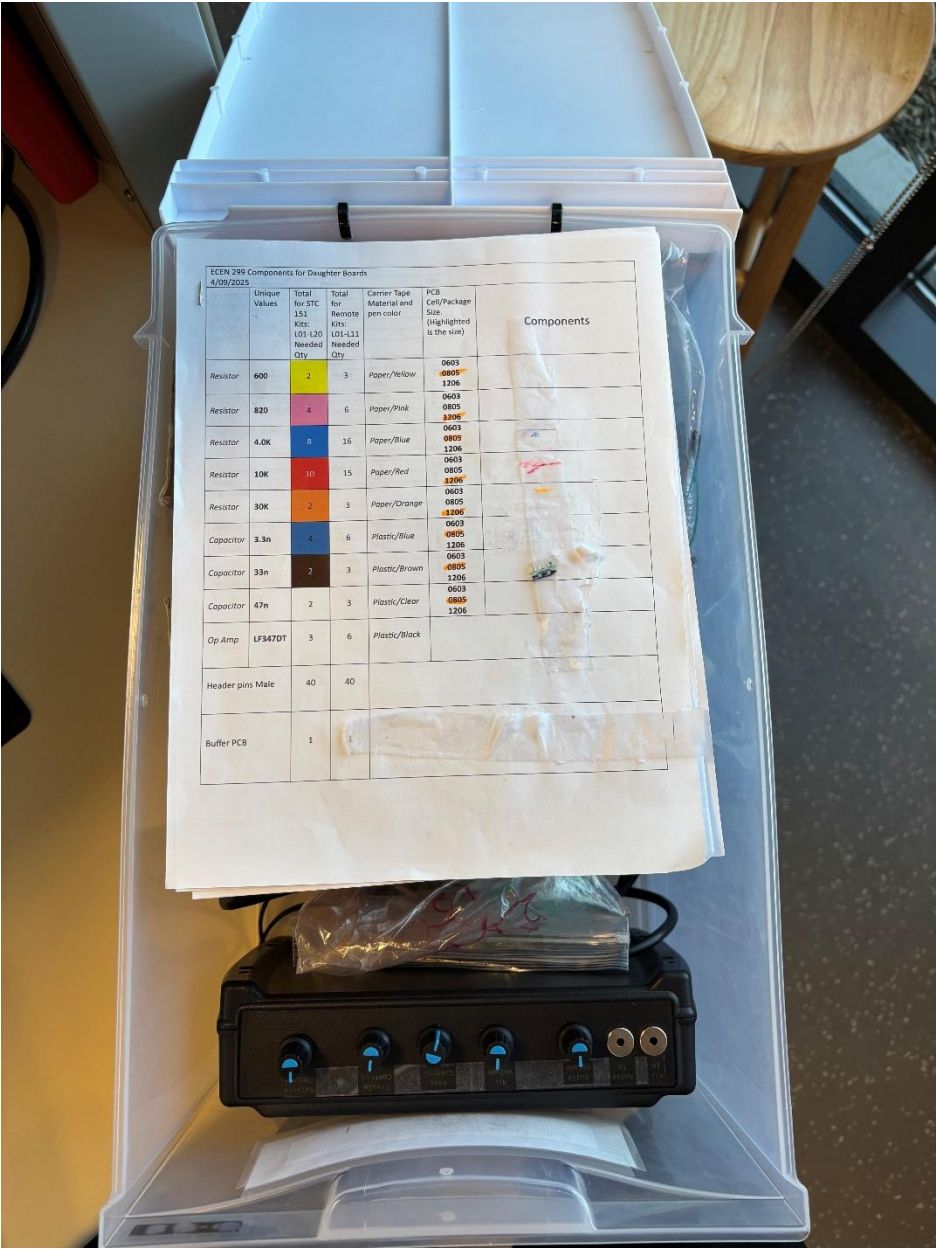
STC 151-15

Figure 10: Bass Right ($a=1$, 20Hz)

Lab Images:







Conclusion

This lab was passed off by Ryan our TA. We found through analysis and demonstration that the daughterboard, and the project, works as intended. Initially we encountered some issues at first in our analysis when trying to demonstrate that the board worked and that it would adjust our Treble and Bass signals audibly. This was due to some improperly placed jumper pins. After this was fixed and adjusted we found it to operate exactly as expected; the audio signals were able to be adjusted for both the Bass and Treble frequencies. We also found on the oscilloscopes that the signals going in to each had the effects we were looking for, where each would attenuate or amplify the signal for each frequency range they were made to do. These boards appear to meet expectations both audibly, and visually from the oscilloscope. Everything fits the design specs except for the 20Hz Bass cut which is barely above the specs but is likely within the tolerance of normal operation. The treble boost and cut at 20kHz are also a little off but also likely within the operational tolerance for normal operation.